ECE2800J

Programming and Elementary Data Structures

Linux

Learning objectives:

Learn how to navigate the directory tree

Learn how to manipulate files/directory

Understand I/O redirection

And a few other useful commands (diff, apt-get…)

Recap from last time

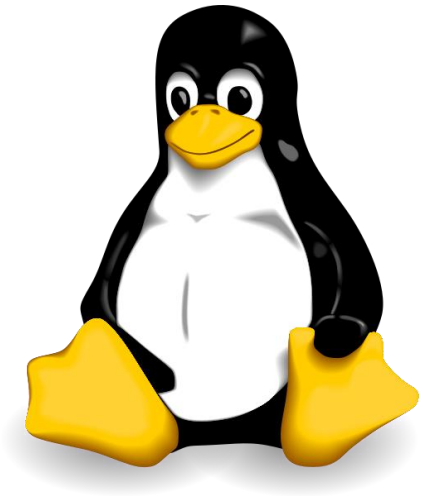
- Course Logistics
- Recitations

Unix

- An **operating system** supporting multitasking and multi-user
- Developed in 1969 by Ken Thompson, Dennis Ritchie, etc. from AT&T Bell Labs
- Many variants (Unix-like OS)
  - Linux
  - BSD (from UC Berkeley)
  - Solaris (from Sun Microsystems)
  - Android (from Google)
  - iOS (from Apple)
  - …

Linux

- A free and open source Unix-like operating system
- First released in 1991 by Linus Torvalds (borned 1969)
- Many distributions
  - Gentoo
  - Red Hat
  - Ubuntu
  - …

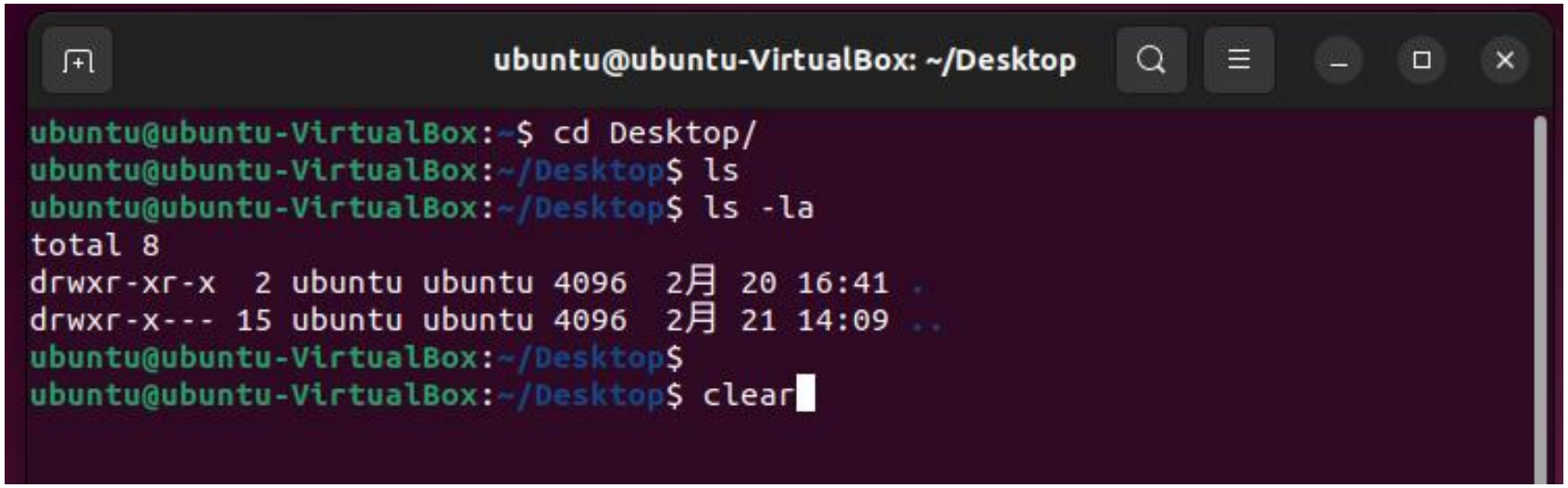


Installing Linux

- Recommended version: **Ubuntu**
  - You can get the .iso file from: http://www.ubuntu.com/download/desktop
  - Suggest to use the latest version.
- Install it directly on your machine
- OR install it on a virtual machine on your Windows/Mac operating system.
  - Install a virtual machine such as VMware Workstation VirtualBox first.
  - VirtualBox can be downloaded at: https://www.virtualbox.org/wiki/Downloads
- WSL (recommended for windows users)

Using Terminal in Linux

- We type commands in the terminal in Linux



- Multiple ways to start a terminal
  - One simple way is to right click and choose from the shortcut menu

Change Directory

- Basic command: cd pathname

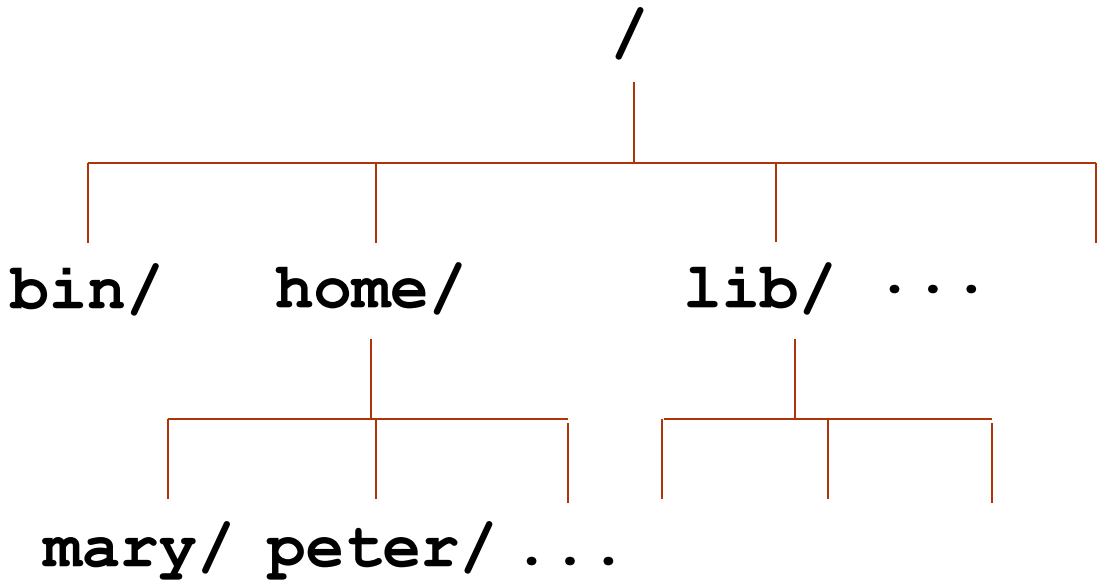
  - E.g., cd /usr/bin

    typical path name format

- Special characters for directories
  - root directory: /
  - home directory: ~
    - Linux is a multi-user operating system. It is your "home directory".
  - current directory: .
  - parent directory: ..

Aside: Root Directory

- Directory in Linux is organized as a tree
- The topmost directory is root directory "/"



List Contents of a Directory

- Basic command: ls directory
  - e.g., ls /home

- ls (i.e., "ls" alone): list the current working directory

Options

what "a" stands for?

- ls -l [directory]: list in long format
- ls -a [directory]: list all files including the hidden files
  - Hidden files: file name begin with a dot, e.g., ".bash_history"
- In Linux, options can be combined together.
  - "ls -la" or "ls -l -a"

Aside: Long Format of File Information

- ls -l

permission		owner	group	file size (in bytes)	modification time	file name
-rw--------	1	john	john	576	Apr 17 1998	weather.txt
drwxr-xr-x	6	john	john	1024	Oct 9 1999	web_page
-rw-rw-r--	1	john	john	276480	Feb 11 20:41	web_site.tar
-rwx-------	1	john	john	5743	Dec 16 1998	my_app

- File permission
  - First character: '-' regular file; 'd' directory
  - Next three: read, write, execution permission of the owner
  - Next three: read, write, execution permission of the group
  - Final three: read, write, execution permission of everyone else

Manipulating Files/Directories

- Create directories: mkdir dir
- Delete directories: rmdir dir
  - Can only remove **empty** directory
- Create an empty file: touch file

Copy Files/Directories

- Basic command: cp source dest

- Variations
  - cp file1 file2: copy the content of file1 into file2
  - cp file1 dir: copy file into a directory
    - cp file1 file2 dir
    - cp file* dir
      - *: wildcard. Can represent any character string (even an empty string!)
  - cp -r dir1 dir2: If dir2 does not exist, copy dir1 as dir2. If dir2 exists, copy dir1 inside dir2

Which Commands List **ALL and Only ALL** Files with the xyz Extension in Current Folder?

Assume no hidden files. Select all the correct answers.

- **A.** `ls ./*xyz`
- **B.** `ls *.xyz`
- **C.** `ls *xyz`
- **D.** None of the above.

Rename/Move a File

- Basic command: mv source dest

- Variations
  - mv file1 file2: rename file1 as file2
  - mv file1 dir: move file into a directory
  - mv dir1 dir2: If dir2 does not exist, then rename dir1 as dir2. If dir2 exists, then move dir1 inside dir2

Delete Files/Directories

- Basic command: rm file

- Variations
  - rm file: delete file
  - rm file1 file2: delete file1 and file2
  - rm -r dir: delete dir along with its contents

- Useful options -i: prompt before every removal
  - To use: alias rm= ‘rm -i’ ;
  - Put it into ~/.bashrc

Edit/Show a File

- Edit file: nano file gedit file
  - advanced editor: vim, emacs
- Show file content
  - cat file
  - less file
    - quit 'less' : press 'q'
    - go to the end: press 'G' (shift + g)
    - go to the beginning: press 'g'
    - search: press '/' , then enter the thing to be searched
    - press 'n' for the next match; press 'N' for the previous match.

I/O Redirection

- Most command line programs display their results on the **standard output**.
  - By default, standard output is our display.

- We can redirect from standard output to a file by using ‘>’.
  - E.g., ls -l > ls_rst.txt: the “ls” result is now in ls_rst.txt

I/O Redirection

- Many commands can accept input from a facility called **standard input**.
  - By default, standard input is our keyboard.

- We can redirect standard input from a file instead of keyboard by using ‘<’.
  - One application: testing
  - E.g., my_add < input.txt
    # my_add is a program taking two inputs from keyboard and output their sum on screen

What does the Following Command Do?

```
sort < fruit.txt > my_favorite.txt
```

Select all the correct answers.

- **A.** The command reads fruit.txt and my_favorite.txt
- **B.** The command reads fruit.txt and writes in my_favorite.txt
- **C.** The elements of fruit.txt are in alphabetic order
- **D.** The elements of my_favorite.txt are in alphabetic order

Other Commands

- Auto completion: type a few characters; then press ‘Tab’
  - If there is a single match, Linux completes the remaining.
  - If there are multiple matches, hit the second time, Linux shows all the possible candidates.

- Compare two files: diff file1 file2
  - If files are the same, no output
  - If there are differences: lines after “<” are from the first file; lines after “>” are from the second file
  - In a summary line: ‘c’ : change; ‘a’ : add; ‘d’ : delete
  - Useful option “-w” : ignore white spaces (space, tab)

Other Commands

- Install a program: sudo apt-get install program
  - E.g., sudo apt-get install emacs
  - sudo command: execute command as a superuser
    - Requires you to type your password
- Remove a program: sudo apt-get autoremove program
- Looking for help? man command e.g., man ls
  - Browse the manual using the same commands as for 'less'

Reference

- http://linuxcommand.org/